

DESIGN AND SIMULATION OF SINGLE INPUT MULTIPLE OUTPUT DC-DC BUCK CONVERTER

*Mini-Project Report submitted to the SASTRA Deemed to be University in partial fulfillment of the
requirements for the award of the degree*

**B. TECH. ELECTRICAL & ELECTRONICS ENGINEERING (SMART GRID &
ELECTRIC VEHICLES)**

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Bonafide Certificate

This is to certify that the report titled **”DESIGN AND SIMULATION OF SINGLE INPUT MULTIPLE OUTPUT DC-DC BUCK CONVERTER”** submitted in partial fulfillment of the requirements for the award of the degree of **B. Tech. Electrical & Electronics Engineering** to the SASTRA Deemed to be University, is a bona-fide record of the work done by **ARUL JEFFRY .A (Reg. No.: 127159008), GURU PRASATH .S.A (Reg. No.: 127159021), VETRIVEL .S (Reg. No.: 127159056)** during the academic year 2025-26, in the School of Electrical & Electronics Engineering, under my supervision. This report has not formed the basis for the award of any degree, diploma, associateship, fellowship or other similar title to any candidate of any University.

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Date : 06/05/2026

Project Viva voce held on _____

Examiner 1

Examiner 2



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Declaration

We declare that the report titled “**DESIGN AND SIMULATION OF SINGLE INPUT MULTIPLE OUTPUT DC-DC BUCK CONVERTER**” submitted by us is an original work done by us under the guidance of **Dr. UMA D, Associate Professor**, School of Electrical & Electronics Engineering, SASTRA Deemed to be University during the Sixth semester of the academic year 2025-26, in the School of Electrical & Electronics Engineering the work is original and wherever we have used materials from other sources, we have given due credit and cited them in the text of the report. This report has not formed the basis for the award of any degree, diploma, associate-ship, fellowship or other similar title to any candidate of any University.

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ABSTRACT

Our work involved detailed simulation analysis of the proposed converter under different operating conditions using MATLAB/Simulink software. The output voltages, inductor currents, switching pulses, and ripple characteristics were analyzed to evaluate converter performance. The additional 3.3V output port was tested for stable voltage regulation and compatibility with low-power electronic loads. Since the project was simulation-based, all system responses were verified through waveform analysis without hardware implementation. The obtained results demonstrated effective closed-loop control operation for the proposed converter system.

Specific Contribution

- Performed simulation testing for the proposed SIMO converter system.
- Analyzed output voltage, current, and switching waveforms.
- Verified the stability and regulation performance of the added 3.3V output.

Specific Learning

- Learned MATLAB/Simulink modeling and simulation techniques.
- Understood waveform analysis of converter circuits.
- Gained practical knowledge in evaluating converter performance parameters.

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ABSTRACT

Our work mainly focused on developing an efficient closed-loop control strategy for maintaining stable output voltages in the multi-output converter system. PI controllers were implemented for each output channel to achieve proper voltage regulation and minimize steady-state error. PWM pulse generation and switching logic were developed using comparator blocks and logical gate operations in MATLAB/Simulink. Special attention was given to the newly added 3.3V output port to ensure proper switching coordination and stable converter operation. The simulation results confirmed that the designed control logic effectively maintained regulated outputs with low ripple characteristics.

Specific Contribution

- Implemented PI-based closed-loop control for all output ports.
- Developed PWM pulse generation and switching logic circuits in MATLAB/Simulink.
- Tuned controller parameters to obtain stable output response with minimum ripple compensation.

Specific Learning

- Learned PWM generation techniques for power electronic converters.
- Understood the working of PI controllers in closed-loop systems.
- Gained experience in logic gate implementation for converter switching control.

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ABSTRACT

Our work presents the simulation of a Single Input Multi-Output (SIMO) DC-DC buck converter with an additional regulated 3.3V output port for low-voltage electronic applications. Unlike the base research work that utilized a photovoltaic source, the proposed system employs a fixed 48V DC supply as the input source for simplified analysis and implementation. The converter topology was developed in MATLAB/Simulink with proper closed-loop voltage regulation using PI controllers. The added 3.3V output was designed to maintain stable voltage under varying load conditions with reduced ripple and improved regulation performance. The complete converter operation and switching sequence were verified through simulation results.

Specific Contribution

- Designed the SIMO converter structure using a 48V DC input source.
- Added an additional regulated 3.3V output port to the existing converter topology.
- Selected suitable converter parameters and verified output regulation through simulation.

Specific Learning

- Learned the operating principle of SIMO DC-DC converters.
- Understood voltage step-down techniques using buck converter topology.
- Gained knowledge in converter parameter selection and output voltage regulation.

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NOTATIONS

ENGLISH SYMBOLS

C	Capacitance
D	Duty Cycle
f_s	Switching Frequency
H	Magnetic Field Intensity
I_{avg}	Average Output Current
L	Inductance
Q	Charge
R	Load Resistance
T_{on}	Switch ON Duration
V_{in}	Input Voltage
V_o	Output Voltage

GREEK SYMBOLS

ΔI_L	Peak-to-peak Inductor Current Ripple
ΔV_o	Peak-to-peak Output Voltage Ripple
Σ	Summation
Ω	Ohm
∞	Infinity

MISCELLANEOUS SYMBOLS

$ x $	Absolute value of x
$\geq$	Greater than or equal to
$\leq$	Less than or equal to
$\approx$	Approximately equal to

ABBREVIATIONS

CCM	Continuous Conduction Mode
DCM	Discontinuous Conduction Mode
MOSFET	Metal Oxide Semiconductor Field Effect Transistor
MPPT	Maximum Power Point Tracking
PI	Proportional–Integral
PID	Proportional–Integral–Derivative
PV	Photovoltaic
PWM	Pulse Width Modulation
SIMO	Single Input Multiple Output
ESR	Equivalent Series Resistance
DC	Direct Current
MCU	Microcontroller Unit

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CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION TO DC–DC CONVERTERS

DC–DC converters are power electronic devices used to convert one level of DC voltage to another level efficiently. They are widely used in applications such as power supplies, electric vehicles, renewable energy systems, and electronic devices. These converters operate using high-frequency switching techniques and energy storage elements like inductors and capacitors to regulate voltage. Depending on the requirement, DC–DC converters can step up (boost), step down (buck), or invert the input voltage. Among these, the buck converter is commonly used for applications requiring lower output voltage from a higher input source.

1.1.1 Single Input Multi-Output (SIMO) Converter

A Single Input Multi-Output (SIMO) converter is a type of DC–DC converter that provides multiple regulated output voltages from a single input source. Unlike conventional systems that use separate converters for each output, a SIMO converter utilizes a shared power stage and control strategy to supply multiple loads. This reduces the number of components, improves efficiency, and minimizes system complexity. SIMO converters are particularly useful in applications where compact size and cost reduction are critical.

1.1.2 Need for Multiple Output Power Supplies

Modern electronic systems require different voltage levels to operate various components such as processors, sensors, communication modules, and power devices. Providing these voltages using separate converters increases system size, cost, and power losses. Therefore, there is a strong need for integrated solutions that can deliver multiple regulated outputs from a single source efficiently. Multi-output converters help in reducing hardware requirements while ensuring proper power distribution across different loads.

1.1.3 Importance of Closed-Loop Control

Closed-loop control is essential in DC–DC converters to maintain a stable output voltage under varying load and input conditions. In open-loop systems, the output voltage can fluctuate due to changes in load or disturbances. By using feedback control techniques such as Proportional–Integral (PI) controllers, the converter continuously compares the output voltage with a reference value and adjusts the duty cycle accordingly. This ensures accurate voltage regulation, improved dynamic response, and reduced steady-state error.

1.1.4 Scope of the Project

The scope of this project includes the design, modeling, and simulation of a SIMO DC–DC buck converter using MATLAB/Simulink. The work focuses on analyzing the converter topology, switching operation, control strategy, and output performance. The system is designed for steady-state under different load conditions. However, the implementation is limited to simulation, and hardware realization is considered as future work.

CHAPTER 2

LITERATURE REVIEW

2.1 PRIOR WORK

”A Bidirectional DC/DC Converter for Fuel Cell Electric Vehicle Driving System” by Huang-Jen Chiu, Hsiu-Ming Huang, Li-Wei Lin, Shann-Chyi Mou, and Pang-Jung Liu was published as a conference paper in the IEEE Vehicle Power and Propulsion Conference (VPPC) in 2008. The paper proposed a bidirectional DC-DC converter for fuel cell electric vehicle applications with improved power transfer capability. Soft-switching techniques were implemented to reduce switching losses and improve converter efficiency. The study demonstrated stable converter operation and effective voltage regulation through simulation and experimental analysis.

“High-Conversion-Ratio Bidirectional DC-DC Converter with Coupled Inductor” by Yi-Ping Hsieh, Jiann-Fuh Chen, Lung-Sheng Yang, Chang-Ying Wu, and Wei-Shih Liu was published in the IEEE Transactions on Industrial Electronics journal in 2014. The paper introduced a coupled inductor-based bidirectional converter capable of achieving high voltage conversion ratio with improved efficiency. Passive clamp circuits were used to reduce switching stress and improve converter performance. The results showed stable power conversion and reduced switching losses under different operating conditions.

“A Novel Investigation on Single-Input Three-Output DC-DC Buck Converter for Electrical Vehicles” was published as a conference paper in the International Conference on Intelligent Sustainable Systems (ICISS) in 2021. The paper presented a SIMO buck converter capable of generating three regulated output voltages from a single input source for electric vehicle applications. PWM switching and closed-loop control techniques were implemented to improve voltage regulation and output stability. Simulation results demonstrated reduced circuit complexity and effective multi-output power conversion performance.

2.2 RESEARCH GAP

Although several SIMO DC-DC converter topologies have been proposed in previous studies, most of them focused mainly on standard output voltage levels such as 5V, 12V, 15V, and 18.5V. Very few works addressed the integration of an additional regulated 3.3V output suitable for low-power embedded systems and modern electronic controllers. Moreover, many existing systems considered photovoltaic sources as the primary input, while limited work was carried out using a fixed 48V DC input source with simplified implementation and control strategies.

Another important limitation identified in previous works is the lack of detailed closed-loop control analysis for additional low-voltage output ports. The coordination of PWM switching logic, PI controller tuning, and ripple minimization for an added 3.3V output has not been sufficiently explored. Therefore, this project focuses on developing a simulation-based SIMO converter with an extra regulated 3.3V output using a 48V DC source, along with proper closed-loop control and switching logic implementation in MATLAB/Simulink.

2.3 PROBLEM STATEMENT

Modern electronic systems require multiple regulated DC voltage levels to power different loads simultaneously. Conventional approaches use separate DC-DC converters for each output, which increases component count, switching losses, system size, and overall cost. Therefore, there is a need for a compact, efficient, and cost-effective solution that can generate multiple regulated output voltages from a single input source while ensuring minimal ripple and stable performance.

2.4 OBJECTIVE

- To design a Single-Input Multi-Output (SIMO) DC–DC buck converter using a 48V DC input source.
- To generate multiple regulated output voltages (18.5V, 15V, 12V, 5V, and 3.3V) for various DC loads.
- To reduce the number of switches and components compared to conventional multi-converter systems.
- To implement closed-loop voltage regulation using Proportional–Integral (PI) controllers and Pulse Width Modulation (PWM) techniques.
- To ensure stable operation with minimal output voltage ripple.
- To design the system for operation in Continuous Conduction Mode (CCM).
- To extend the converter functionality by adding a 3.3V output for low-power embedded applications.

CHAPTER 3

DERIVED SYSTEM

3.1 OVERVIEW OF THE DERIVED SYSTEM

The derived system is a Single Input Multi-Output (SIMO) DC-DC buck converter designed to generate multiple regulated DC output voltages from a single 48V DC input source. Unlike the previous work, which uses a photovoltaic (PV) source, this project utilizes a fixed DC supply for simplified implementation and simulation analysis. An additional regulated 3.3V output port is incorporated into the converter to support low-voltage electronic devices such as microcontrollers, sensors, and embedded systems.

The converter consists of semiconductor switches, inductors, capacitors, PWM pulse generation circuits, and closed-loop PI controllers. The buck converter topology is used to step down the 48V input into different output voltage levels. Proper switching logic and PWM control techniques are implemented in MATLAB/Simulink to achieve stable converter operation and avoid prohibited switching conditions. The PI controllers continuously regulate the output voltages by adjusting the duty cycle based on the error between reference and actual output voltages.

The complete system is designed and analyzed using MATLAB/Simulink software. Simulation results such as output voltages, switching pulses, inductor voltage are studied to evaluate converter performance. The derived system successfully maintains stable multiple output voltages with low ripple and proper closed-loop control operation. Since the project is simulation-based, all converter characteristics and performance parameters are verified through waveform analysis without hardware implementation.

3.2 SYSTEM STRUCTURE

The Derived system structure consists of a Single Input Multi-Output (SIMO) DC-DC buck converter supplied by a 48V DC input source to generate multiple regulated output voltages. The system includes semiconductor switches, inductors, capacitors, PWM pulse generation circuits, and closed-loop PI controllers for stable converter operation.

An additional regulated 3.3V output port is integrated to support low-voltage electronic applications. PWM switching and logical gate operations are used to control the switching sequence and maintain proper voltage regulation. The block diagram of our derived system is shown in fig 3.1

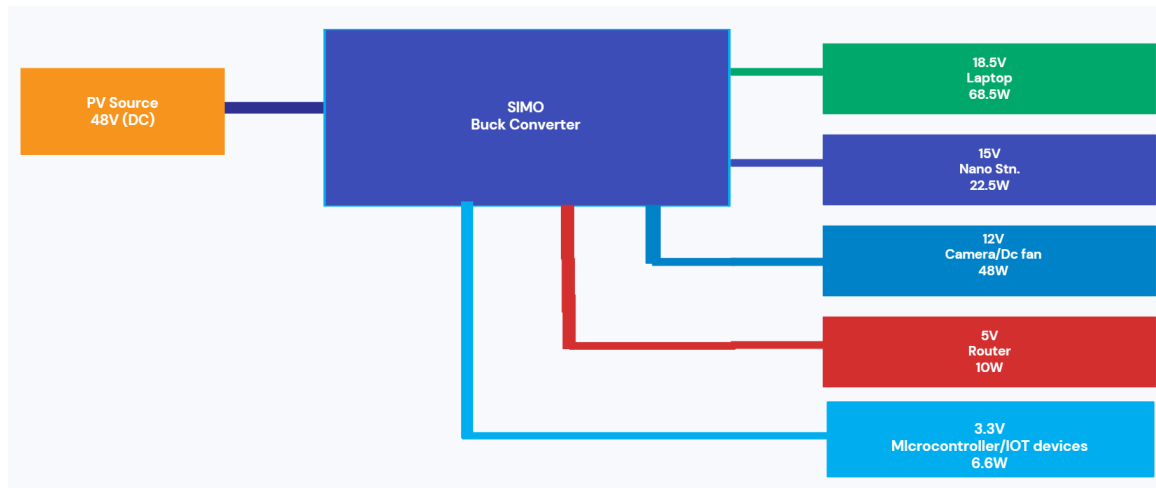


Figure 3.1: Block Diagram of Derived System

Circuit Diagram of Derived System

The below fig 3.2 shows our derived system circuit diagram, that has been constructed. The circuit diagram consists of one DC voltage source for input, 6 Mosfet's for switching purposes, 5 inductor for energy storage and voltage conversion, 5 capacitor for filtering and lastly 6 load resistors.

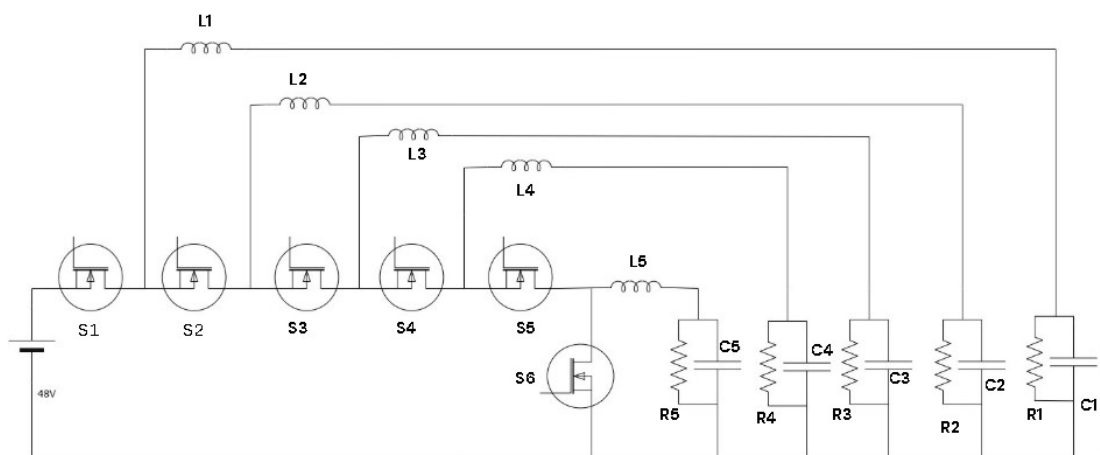


Figure 3.2: Circuit Diagram of Derived System

3.3 ADDITION OF 3.3V OUTPUT PORT

The addition of the 3.3V output port is mainly intended to support low-voltage electronic devices such as microcontrollers, sensors, IoT modules, and embedded systems. Most existing SIMO converter designs mainly focus on higher output voltages like 5V, 12V, and 15V, which are not directly suitable for many modern digital electronics. By integrating a dedicated 3.3V output within the same converter topology, the need for additional voltage regulator circuits is reduced. This helps in simplifying the overall converter structure and improving system integration.

Another important reason for adding the 3.3V output is to improve converter flexibility and efficiency. Instead of converting higher voltages to 3.3V using separate regulators, the derived system directly generates the required low voltage from the 48V DC input source. The closed-loop PI controller maintains stable voltage regulation and minimizes output ripple under varying load conditions. This additional output makes the converter more suitable for multi-load applications requiring both high-voltage and low-voltage DC supplies.

3.4 WORKING PRINCIPLE OF DERIVED SYSTEM

The derived system operates on the principle of a Single Input Multi-Output (SIMO) DC-DC buck converter, where a single 48V DC input source is converted into multiple regulated lower DC output voltages. The converter uses semiconductor switches, inductors, and capacitors to step down the input voltage to the required output levels including the additional 3.3V output port. The switching operation is controlled using PWM pulses generated at a fixed switching frequency.

During converter operation, the semiconductor switches turn ON and OFF according to the PWM control signals. When a switch is turned ON, energy from the input source is stored in the corresponding inductor, and when the switch is turned OFF, the stored energy is transferred to the load through the output filter capacitor. This charging and discharging action of inductors produces the required regulated output voltages for different DC loads.

Closed-loop PI controllers are used to maintain stable output voltages and reduce ripple content. The controller continuously compares the reference voltage with the actual

output voltage and generates an error signal. Based on this error, the PWM duty cycle is automatically adjusted to achieve proper voltage regulation. Logical gate operations are also implemented to avoid prohibited switching states and ensure safe and reliable converter operation.

3.5 MODE OF OPERATION

State topology	TS1	TS2	TS3	TS4	TS5	TS6
Switch1	1	1	1	1	1	0
Switch2	1	1	1	1	0	1
Switch3	1	1	1	0	1	1
Switch4	1	1	0	1	1	1
Switch5	1	0	1	1	1	1
Switch6	0	1	1	1	1	1

Table 3.1: STATE TOPOLOGY

The above table 3.1 shows the Mosfet's switching operation, like when the switch is need to be open and close.

Mode-1 (S6 - OFF)

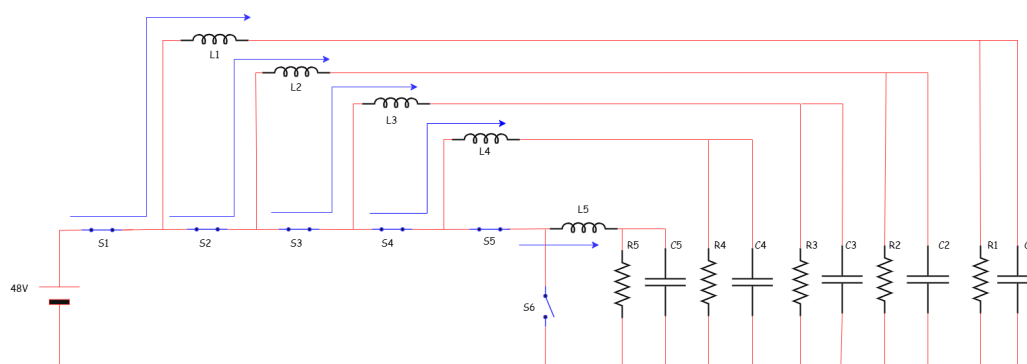


Figure 3.3: S6 - OFF

The above fig 3.3 shows the flow of current, when the S6 is opened.

- The current flows through all inductors.
- All the inductors get charged.
- Current flows through load resistor and the filter capacitor accordingly.

Mode-2 (S5 - OFF)

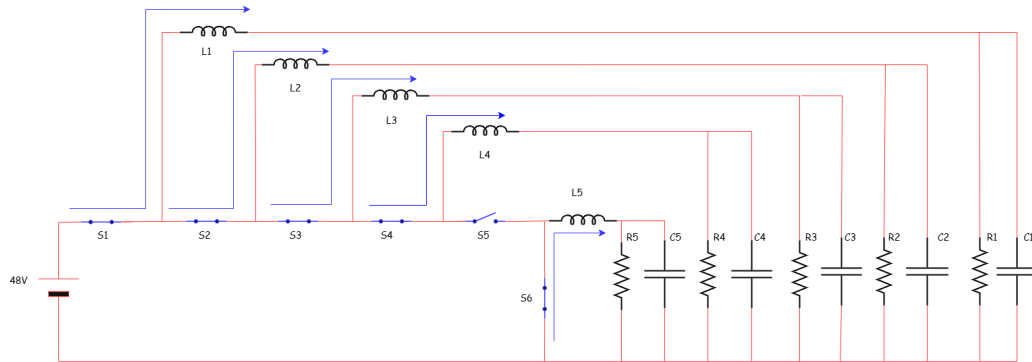


Figure 3.4: S5 - OFF

The above fig 3.4 shows the flow of current, when the S5 is opened.

- The current flows through inductors(L1,L2,L3,L4).
- The inductors(L1-L4) gets charged, the inductor L5 is getting discharged.
- Inductor current flows through the C5,R5.

Mode-3 (S4 - OFF)

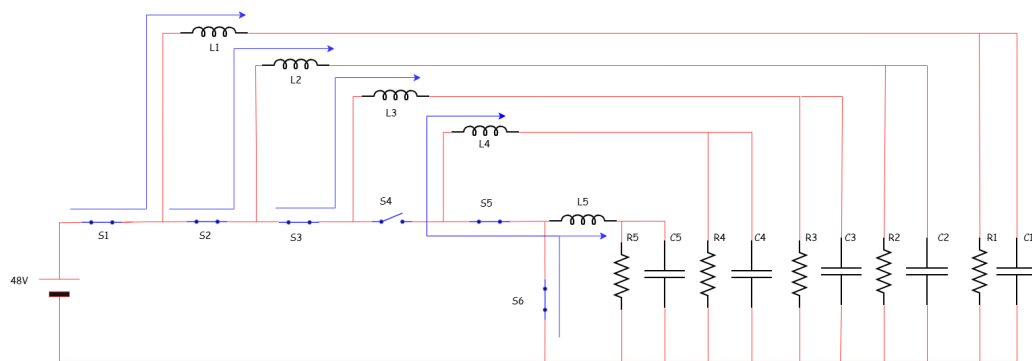


Figure 3.5: S4 - OFF

The above fig 3.5 shows the flow of current, when the S4 is opened.

- The current flows through inductors(L1,L2,L3).
- The inductors(L1,L2,L3) gets charged, the inductos L5,L4 is getting discharged.
- Inductor current flows through (C5,R5 and C4,R4).

Mode-4 (S3 - OFF)

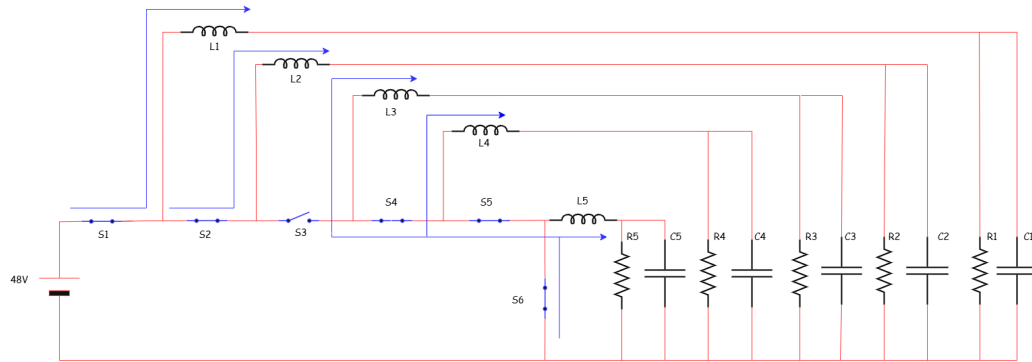


Figure 3.6: S3 - OFF

The above fig 3.6 shows the flow of current, when the S3 is opened.

- The current flows through inductors(L1,L2).
- The inductors(L1,L2) gets charged, the inductos L5,L4,L3 is getting discharged.
- Inductor current flows through (C5,R5 and C4,R4 and C3,R4).

Mode-5 (S2 - OFF)

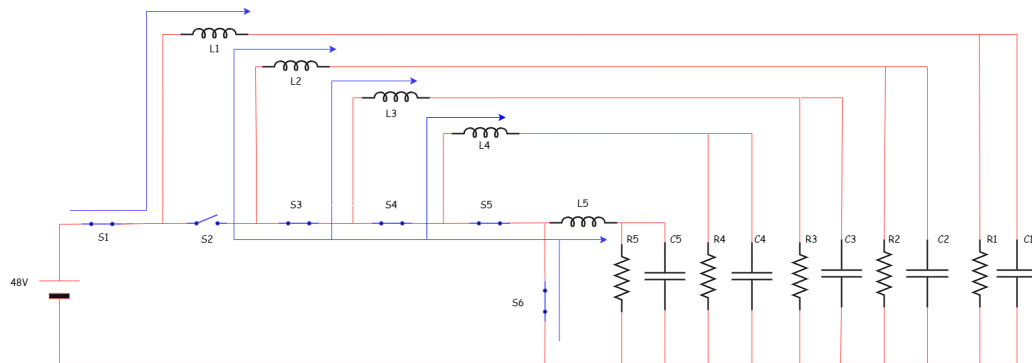


Figure 3.7: S2 - OFF

The above fig 3.7 shows the flow of current, when the S2 is opened.

- The current flows through inductor(L1).
- The inductor(L1) gets charged, the inductos (L2,L3,L4,L5) is getting discharged.
- Inductor current flows through (C5,R5 and C4,R4 and C3,R3 and C2,R2).

Mode-6 (S1 - OFF)

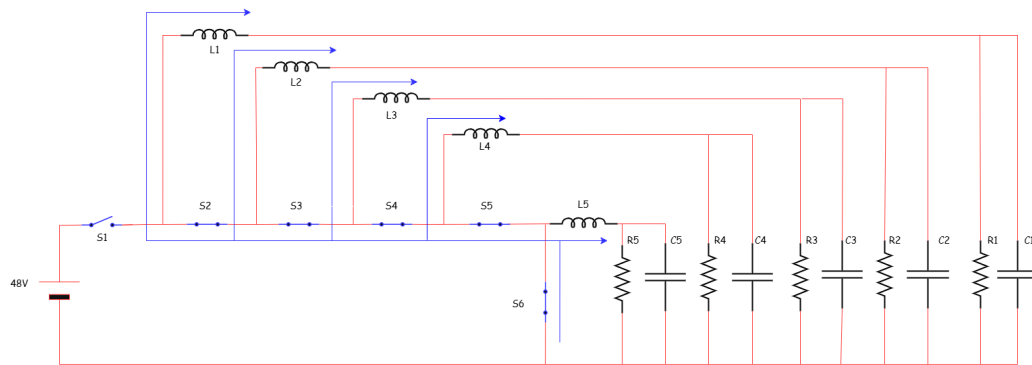


Figure 3.8: S1 - OFF

The above fig 3.8 shows the flow of current, when the S1 is opened.

- Inductor current flows through (C5,R5 and C4,R4 and C3,R3 and C2,R2 and C1,R1).
- All inductors get discharged.

3.6 ADVANTAGE OF DERIVED SYSTEM

The derived Single Input Multi-Output (SIMO) DC-DC converter provides multiple regulated output voltages from a single 48V DC input source, which reduces the need for separate converters for each load. The addition of the regulated 3.3V output port makes the system suitable for powering low-voltage embedded and electronic devices along with higher voltage loads. The use of closed-loop PI controllers improves voltage regulation, reduces output ripple, and ensures stable converter operation under varying load conditions.

The derived system also reduces component count and overall circuit complexity by using a single converter topology for multiple outputs. PWM-based switching and logical gate control techniques help in achieving proper switching operation and avoiding prohibited states. Since the entire system is implemented in MATLAB/Simulink, the converter performance can be analyzed effectively through waveform observation and simulation testing before hardware implementation.

CHAPTER 4

DESIGN AND ANALYSIS

4.1 DESIGN SPECIFICATIONS

The derived SIMO DC–DC buck converter is designed with the following system-level and per-output specifications.

Table 4.1: System-level design specifications

Parameter	Symbol	Value
Input voltage	V_{in}	48 V
Switching frequency	f_s	10 kHz
Output voltage ripple	ΔV_o	$\leq 0.02\%$ of V_o
Operating mode	—	Continuous Conduction Mode (CCM)
Number of outputs	—	5
Number of switches	—	6 (S1–S6)
Control strategy	—	PI controller

The above table 4.1 shows the design specifications, we need for the calculation of the inductors and capacitors.

Table 4.2: Per-output port specifications

Output	V_o (V)	R (Ω)	I_{avg} (A)	Power (W)	Load device
Out 1	18.5	5.0	3.70	68.45	Laptop charger
Out 2	15.0	10.0	1.50	22.50	Nano station
Out 3	12.0	3.0	4.00	48.00	Camera / DC fan
Out 4	5.0	2.5	2.00	10.00	Router / Mobile
Out 5	3.3	1.65	2.00	6.60	Microcontroller / IoT device

The above table 4.2 shows the output port specifications, where the output voltage and load of the devices are assumed for the applications.

4.2 BUCK CONVERTER DESIGN EQUATIONS

A buck (step-down) converter produces a lower DC output voltage than its input while maintaining the same polarity. The converter operates in Continuous Conduction Mode (CCM) throughout this design.

4.2.1 Output Voltage Equation

In buck converter, the output voltage depends on the duty cycle of the switch and the input voltage.

$$V_o = D \cdot V_{in} \quad (4.1)$$

where D is the duty cycle of the corresponding switch ($0 < D < 1$). Rearranging for the duty cycle:

$$D = \frac{V_o}{V_{in}} \quad (4.2)$$

Each of the five outputs has an independent duty cycle D_k ($k = 1, 2, 3, 4, 5$) controlled by switches S_1 through S_5 respectively. Switch S_6 acts as a shared freewheeling switch and is driven by the NAND combination of all other gate signals to prevent forbidden states, which may short circuit the converter.

4.2.2 Inductor Design Equation

During the ON interval $T_{on} = D/f_s$, the voltage across the inductor is $(V_{in} - V_o)$. Applying the inductor volt-second relationship $v_L = L \frac{di_L}{dt}$:

$$L \geq \frac{V_o (1 - D)}{f_s \cdot \Delta I_L} \quad (4.3)$$

where ΔI_L is the peak-to-peak inductor current ripple. A lower ripple current requires a larger inductance. In this design, an absolute ripple current target of approximately 4.5 mA is maintained across all outputs to keep component selection consistent.

4.2.3 Capacitor Design Equation

The output capacitor filters the triangular ripple current from the inductor. Assuming a symmetric triangular ripple waveform, charge balance over one switching cycle gives:

$$C \geq \frac{\Delta I_L}{8 f_s \Delta V_o} \quad (4.4)$$

where ΔV_o is the allowable peak-to-peak output voltage ripple. Based on the design specification of $\leq 0.02\%$ ripple:

$$\Delta V_o = 0.0002 \times V_o \quad (4.5)$$

A lower output voltage results in a tighter absolute ripple budget, which demands a larger capacitor. Consequently, the lowest-voltage output requires the largest capacitor in the system.

4.3 DUTY CYCLE CALCULATION

Using Eq. (4.2) and the specifications from Section 4.1, the duty cycle for each output is calculated as follows.

Output 1 — $V_{o1} = 18.5 \text{ V}$

$$D_1 = \frac{V_{o1}}{V_{in}} = \frac{18.5}{48} = 0.3854 \quad (4.6)$$

Output 2 — $V_{o2} = 15 \text{ V}$

$$D_2 = \frac{V_{o2}}{V_{in}} = \frac{15}{48} = 0.3125 \quad (4.7)$$

Output 3 — $V_{o3} = 12 \text{ V}$

$$D_3 = \frac{V_{o3}}{V_{in}} = \frac{12}{48} = 0.2500 \quad (4.8)$$

Output 4 — $V_{o4} = 5 \text{ V}$

$$D_4 = \frac{V_{o4}}{V_{in}} = \frac{5}{48} = 0.1042 \quad (4.9)$$

Output 5 — $V_{o5} = 3.3 \text{ V}$

$$D_5 = \frac{V_{o5}}{V_{in}} = \frac{3.3}{48} = 0.0688 \quad (4.10)$$

The above table 4.3 shows the duty cycle values corresponding to the output voltage. The output voltage decreases, the duty cycle decreases and $(1 - D)$ increases. This means the

Table 4.3: Summary of duty cycle values

Output	V_o (V)	D	$(1 - D)$
Out 1	18.5	0.3854	0.6146
Out 2	15.0	0.3125	0.6875
Out 3	12.0	0.2500	0.7500
Out 4	5.0	0.1042	0.8958
Out 5	3.3	0.0688	0.9312

inductor is in the freewheeling state for a longer fraction of each cycle, which directly influences the capacitor sizing in Sections 4.4 and 4.5.

4.4 INDUCTOR CALCULATION

Using Eq. (4.3) with $f_s = 10$ kHz and $\Delta I_L \approx 5$ mA for each output:

Output 1 — $V_{o1} = 18.5$ V, $R_1 = 5$ Ω

$$L_1 \geq \frac{V_{o1} (1 - D_1)}{f_s \cdot \Delta I_{L1}} = \frac{18.5 \times 0.6146}{10000 \times 0.0045} = \frac{11.370}{45} = 252.7 \text{ mH} \approx \boxed{280 \text{ mH}} \quad (4.11)$$

Output 2 — $V_{o2} = 15$ V, $R_2 = 10$ Ω

$$L_2 \geq \frac{V_{o2} (1 - D_2)}{f_s \cdot \Delta I_{L2}} = \frac{15 \times 0.6875}{10000 \times 0.0045} = \frac{10.3125}{45} = 229.2 \text{ mH} \approx \boxed{230 \text{ mH}} \quad (4.12)$$

Output 3 — $V_{o3} = 12$ V, $R_3 = 3$ Ω

$$L_3 \geq \frac{V_{o3} (1 - D_3)}{f_s \cdot \Delta I_{L3}} = \frac{12 \times 0.7500}{10000 \times 0.005} = \frac{9.000}{50} = \boxed{180 \text{ mH}} \quad (4.13)$$

Output 4 — $V_{o4} = 5$ V, $R_4 = 2.5$ Ω

$$L_4 \geq \frac{V_{o4} (1 - D_4)}{f_s \cdot \Delta I_{L4}} = \frac{5 \times 0.8958}{10000 \times 0.005} = \frac{4.479}{50} = 89.6 \text{ mH} \approx \boxed{90 \text{ mH}} \quad (4.14)$$

Output 5 — $V_{o5} = 3.3 \text{ V}$, $R_5 = 1.65 \Omega$

$$L_5 \geq \frac{V_{o5} (1 - D_5)}{f_s \cdot \Delta I_{L5}} = \frac{3.3 \times 0.9312}{10000 \times 0.005} = \frac{3.073}{50} = 61.5 \text{ mH} \approx \boxed{62 \text{ mH}} \quad (4.15)$$

Table 4.4: Summary of inductor values

Output	V_o	D	Calculated	Selected
L_1	18.5 V	0.3854	252.7 mH	280 mH
L_2	15.0 V	0.3125	229.2 mH	230 mH
L_3	12.0 V	0.2500	180.0 mH	180 mH
L_4	5.0 V	0.1042	89.6 mH	90 mH
L_5	3.3 V	0.0688	61.5 mH	62 mH

The above table 4.4 shows the inductors values that have been calculated and the inductor have been selected. The selected inductor values are more or less commonly used inductor values.

4.5 CAPACITOR CALCULATION

Output 1 — $V_{o1} = 18.5 \text{ V}$

$$\Delta V_{o1} = 0.0002 \times 18.5 = 3.70 \text{ mV} \quad (4.16)$$

$$C_1 \geq \frac{\Delta I_{L1}}{8 f_s \Delta V_{o1}} = \frac{0.00406}{8 \times 10000 \times 0.0037} = \frac{0.00406}{296} = 13.7 \text{ } \mu\text{F} \approx \boxed{35 \text{ } \mu\text{F}} \quad (4.17)$$

Output 2 — $V_{o2} = 15 \text{ V}$

$$\Delta V_{o2} = 0.0002 \times 15 = 3.00 \text{ mV} \quad (4.18)$$

$$C_2 \geq \frac{\Delta I_{L2}}{8 f_s \Delta V_{o2}} = \frac{0.00448}{8 \times 10000 \times 0.003} = \frac{0.00448}{240} = 18.7 \text{ } \mu\text{F} \approx \boxed{43 \text{ } \mu\text{F}} \quad (4.19)$$

Output 3 — $V_{o3} = 12 \text{ V}$

$$\Delta V_{o3} = 0.0002 \times 12 = 2.40 \text{ mV} \quad (4.20)$$

$$C_3 \geq \frac{\Delta I_{L3}}{8 f_s \Delta V_{o3}} = \frac{0.005}{8 \times 10000 \times 0.0024} = \frac{0.005}{192} = 26.0 \text{ } \mu\text{F} \approx \boxed{54 \text{ } \mu\text{F}} \quad (4.21)$$

Output 4 — $V_{o4} = 5 \text{ V}$

$$\Delta V_{o4} = 0.0002 \times 5 = 1.00 \text{ mV} \quad (4.22)$$

$$C_4 \geq \frac{\Delta I_{L4}}{8 f_s \Delta V_{o4}} = \frac{0.00498}{8 \times 10000 \times 0.001} = \frac{0.00498}{80} = 61.3 \text{ } \mu\text{F} \approx \boxed{62 \text{ } \mu\text{F}} \quad (4.23)$$

Output 5 — $V_{o5} = 3.3 \text{ V}$

$$\Delta V_{o5} = 0.0002 \times 3.3 = 0.66 \text{ mV} \quad (4.24)$$

$$C_5 \geq \frac{\Delta I_{L5}}{8 f_s \Delta V_{o5}} = \frac{0.00496}{8 \times 10000 \times 0.00066} = \frac{0.00496}{52.8} = 93.9 \text{ } \mu\text{F} \approx \boxed{100 \text{ } \mu\text{F}} \quad (4.25)$$

Table 4.5: Summary of capacitor values

Output	V_o	ΔV_o (mV)	ΔI_L (mA)	Calculated	Selected
C_1	18.5 V	3.70	4.06	13.7 μF	35 μF
C_2	15.0 V	3.00	4.48	18.7 μF	43 μF
C_3	12.0 V	2.40	5.00	26.0 μF	54 μF
C_4	5.0 V	1.00	4.98	61.3 μF	62 μF
C_5	3.3 V	0.66	4.96	93.9 μF	100 μF

By observing the above table 4.5, the capacitor value increases as the output voltage decreases. This is because a lower V_o results in a smaller absolute ripple budget ΔV_o , requiring a larger capacitor to satisfy the same 0.02% ripple specification.

4.6 SELECTION OF COMPONENT VALUES

4.6.1 Inductor Selection

For Out 2 through Out 5, the selected values are obtained by rounding the calculated value upward to the nearest standard preferred component value. Rounding is always preferred since a slightly larger inductance reduces ΔI_L below the design target, which only improves ripple performance. Out 3 (12 V) and Out 4 (5 V) require no rounding as the calculated values of 180 mH and 89.6 mH coincide directly with standard values.

Out 1 (18.5 V) is the only output with a deliberate overdesign, where the selected value of 270 mH exceeds the calculated 252.7 mH by approximately 11%. This rail supplies the laptop charger, which is the most dynamically demanding load. A larger inductance reduces the rate of current change $di/dt = V/L$ during load transients, and ensures the converter remains in continuous conduction mode (CCM) across the full load range, preventing instability at light load.

4.6.2 Capacitor Selection

The selected capacitor values for Out 1 through Out 3 are higher than the calculated minimum. The design formula given in Eq. (4.4) represents the minimum capacitance required to meet the ripple specification under ideal conditions; in practice, a value greater than this minimum must be selected for the following reasons.

The 0.02% ripple specification applies to each output voltage individually. Since the absolute ripple budget $\Delta V_o = 0.0002 \times V_o$ becomes smaller as V_o decreases, the formula inherently demands a larger capacitance for lower output voltages, and the selected value must satisfy this tighter constraint.

Practical capacitors are available only in standard preferred values (E6, E12, E24 series). The calculated minimum is therefore always rounded up to the nearest available standard capacitor value to ensure the ripple specification is not violated.

The design formula yields a theoretical minimum. To guarantee the ripple specification is met with margin across component tolerances and operating conditions, the selected capacitance must always be greater than or equal to the calculated value.

CHAPTER 5

SIMULATION RESULTS

5.1 MATLAB/SIMULINK MODEL:

The below figure 5.1 shows the simulation of the derived system. This model includes the gate logic subsystem and closed loop subsystem.

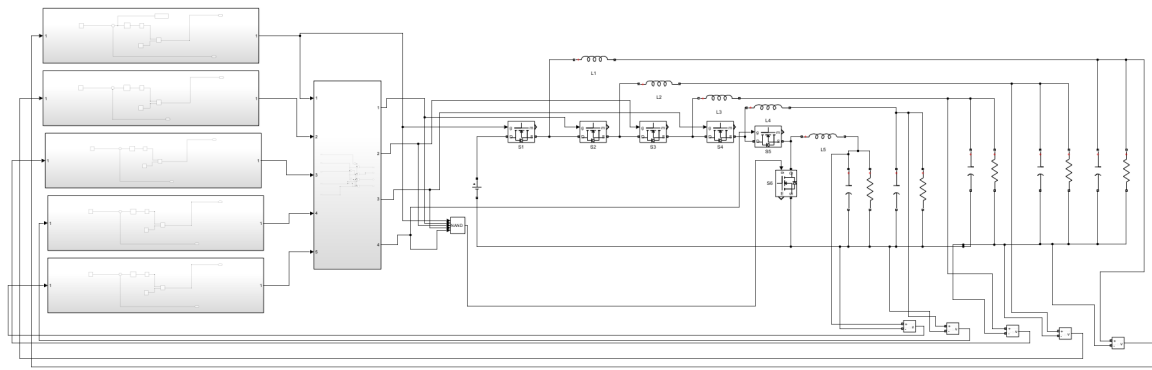


Figure 5.1: Simulation

5.2 GATE LOGIC SUBSYSTEM:

The below figure 5.2 shows the subsystem of the gate logic in our derived system.

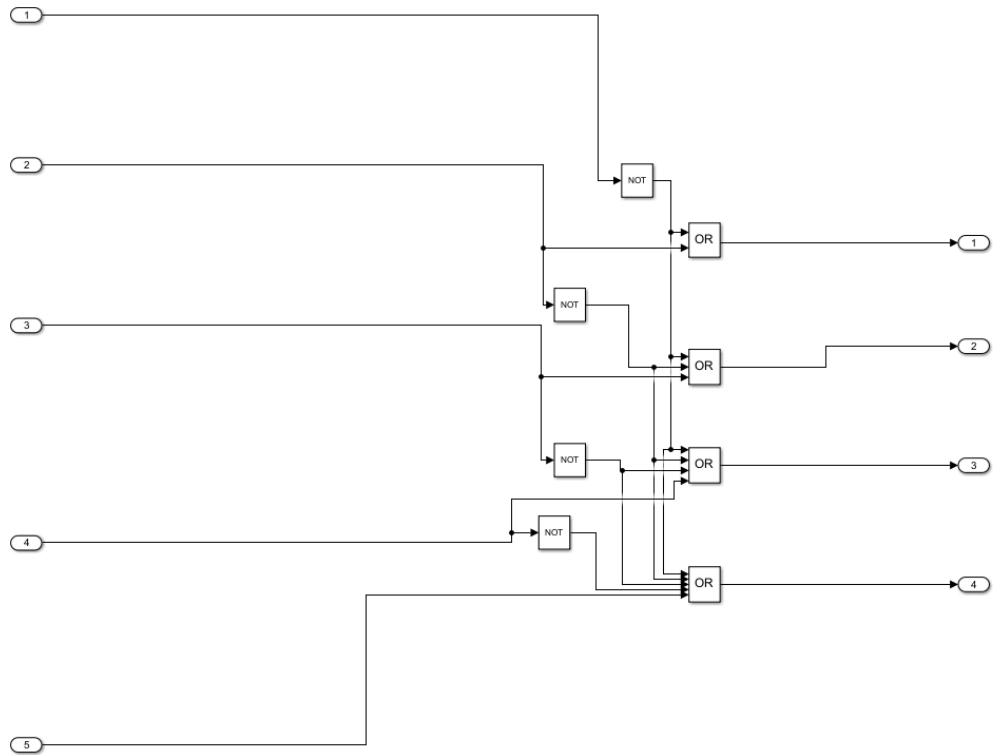


Figure 5.2: Gate logic subsystem

5.3 OUTPUT WAVEFORM FOR 18.5V

The below figure 5.3 shows the output waveform of 18.5V. The regulated output voltage can be used to supply the loads such as a laptop charger.

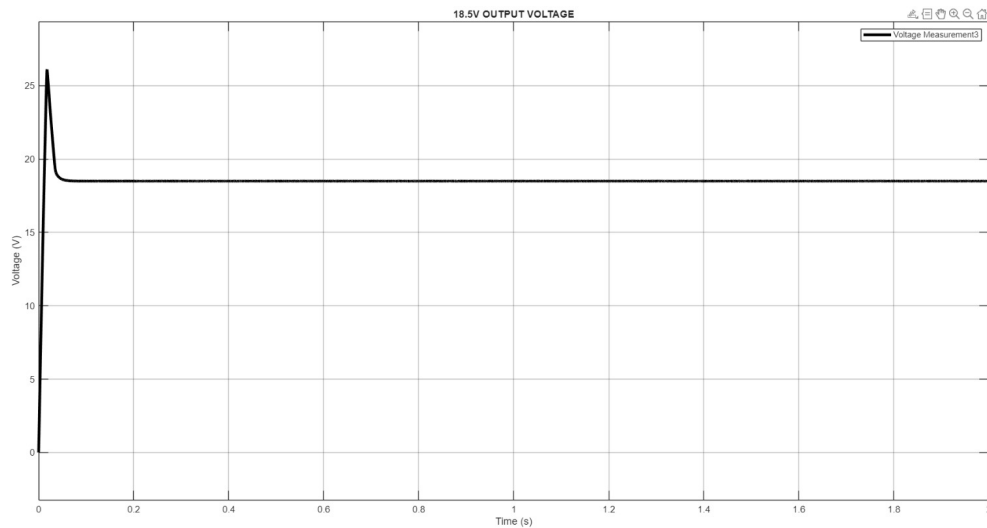


Figure 5.3: Output Waveform for 18.5V

5.4 OUTPUT WAVEFORM FOR 15V

The below figure 5.4 shows the output waveform of 15V. The regulated output voltage can be used to supply the loads such as a nano-station.

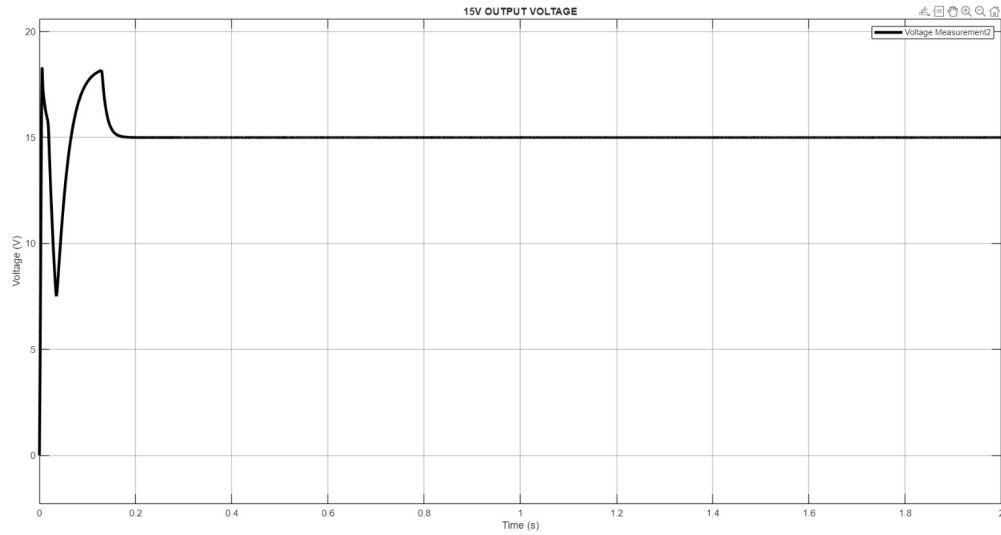


Figure 5.4: Output Waveform for 15V

5.5 OUTPUT WAVEFORM FOR 12V

The below figure 5.5 shows the output waveform of 12V. The regulated output voltage can be used to supply the loads such as a camera/DC fan.

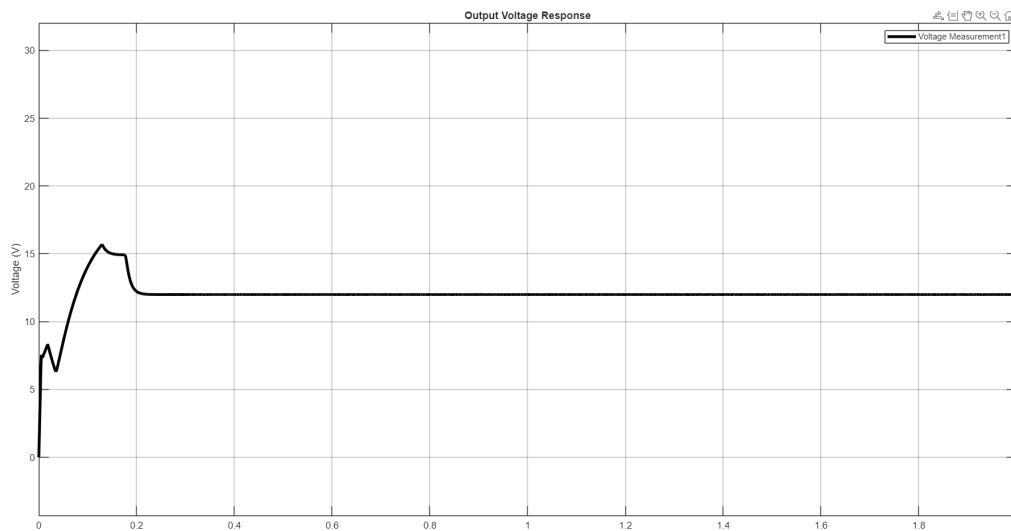


Figure 5.5: Output Waveform for 12V

5.6 OUTPUT WAVEFORM FOR 5V

The below figure 5.6 shows the output waveform of 5V. The regulated output voltage can be used to supply the loads such as a router/mobile.

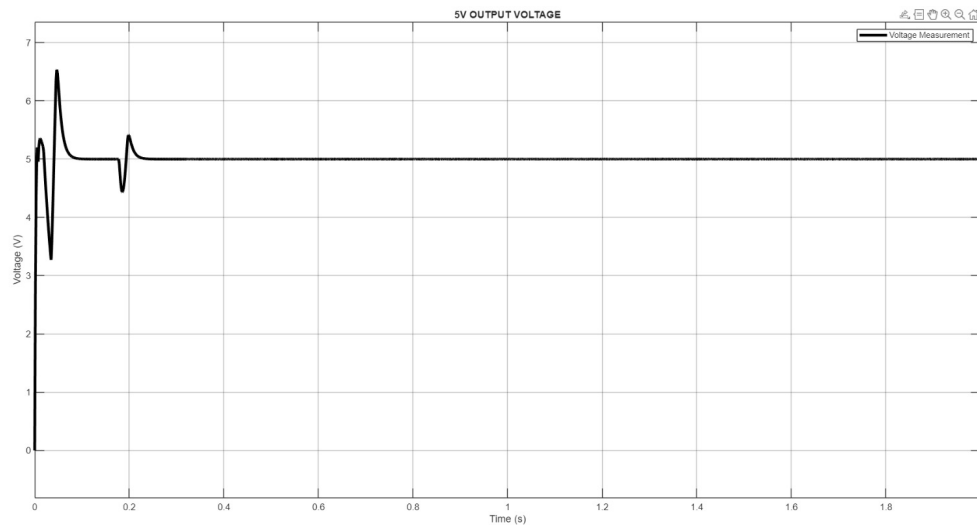


Figure 5.6: Output Waveform for 5V

5.7 OUTPUT WAVEFORM FOR 3.3V

The below figure 5.7 shows the output waveform of 3.3V. The regulated output voltage can be used to supply the loads such as a microcontroller/IOT devices.

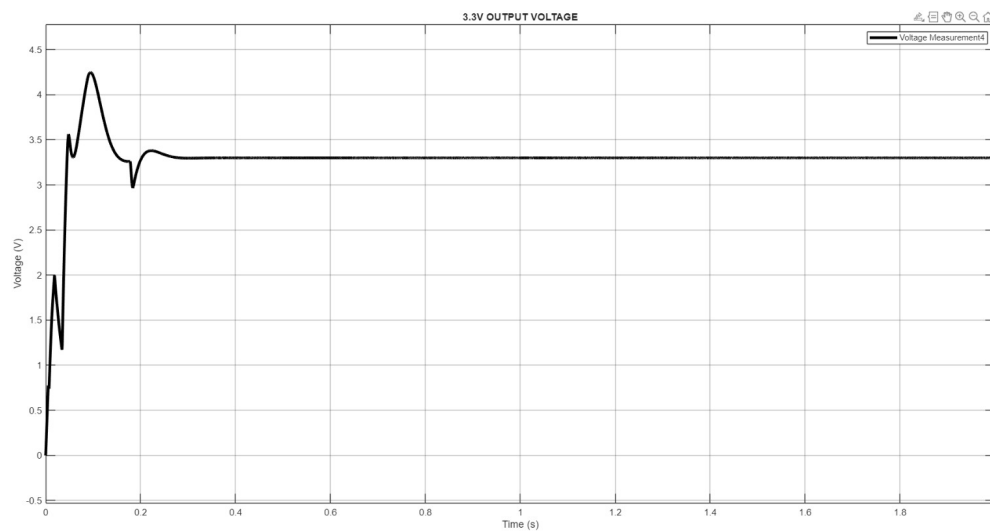


Figure 5.7: Output Waveform for 3.3V

5.8 INDUCTOR VOLTAGE WAVEFORM

The below figure 5.8 shows the inductor voltage waveform. This waveform indicates the charging and discharging of the inductors.

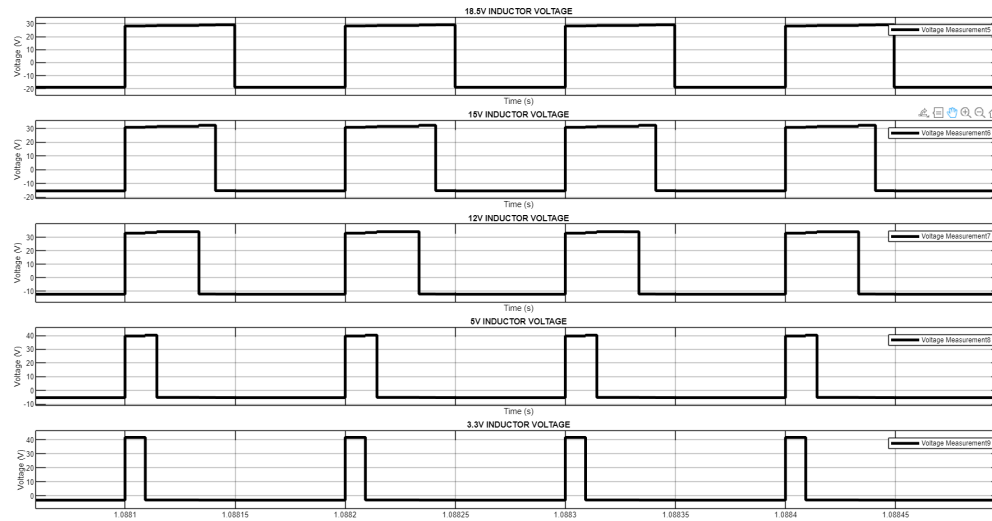


Figure 5.8: Inductor voltage waveform

5.9 WAVEFORM OF SWITCHES

The below figure 5.9 shows the waveform of switches. This waveform indicates the ON and OFF state of the MOSFET's.

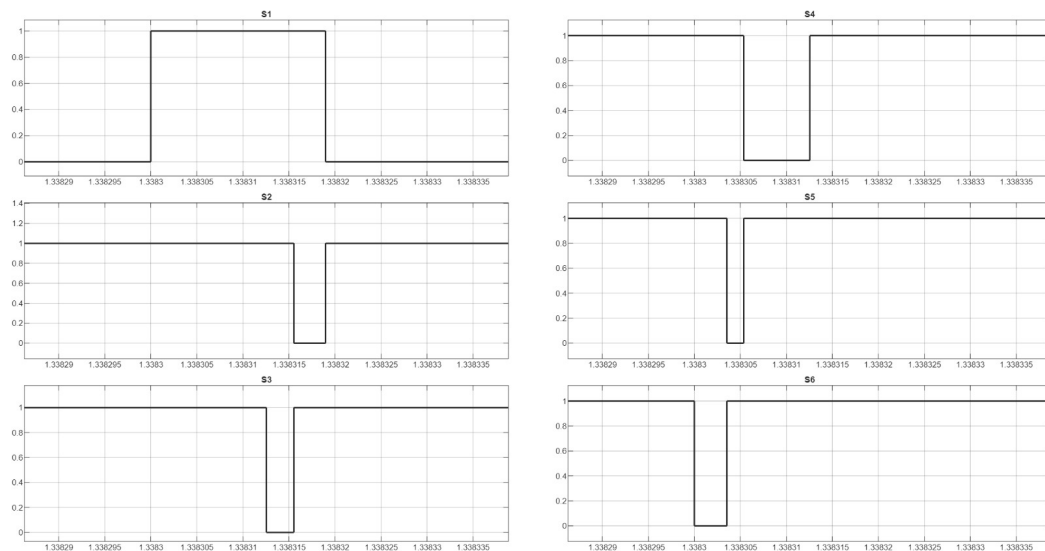


Figure 5.9: Waveform of switches

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 CONCLUSION

This project presented the design and simulation of a Single Input Multi-Output (SIMO) DC-DC buck converter capable of generating multiple regulated output voltages from a single 48V DC input source. An additional regulated 3.3V output port was successfully integrated to support low-voltage electronic applications such as embedded systems and sensors.

Closed-loop PI controllers and PWM switching techniques were implemented to achieve stable output voltage regulation with reduced ripple characteristics. The complete converter system was modeled and analyzed using MATLAB/Simulink software.

Simulation results verified the proper operation of the converter under different conditions. Output voltage waveforms, switching pulses, and inductor voltage characteristics confirmed stable performance and effective closed-loop control.

The system demonstrated improved flexibility, reduced circuit complexity, and the ability to supply multiple DC loads using a single converter topology. Since the project was simulation-based, performance was validated through waveform analysis.

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6.2 FUTURE SCOPE

In future work, the proposed converter system can be integrated with a photovoltaic (PV) source instead of a fixed 48V DC supply to support renewable energy applications. MPPT (Maximum Power Point Tracking) techniques can also be implemented to improve the efficiency of solar energy utilization and enhance converter performance under varying environmental conditions.

The system can further be extended with hardware implementation using digital controllers such as microcontrollers or DSP-based control platforms. Advanced control techniques, improved switching methods, and protection circuits can also be incorporated to reduce switching losses, improve efficiency, and support high-power practical applications such as electric vehicles, battery charging systems, and IoT-based power supply systems.

CHAPTER 7

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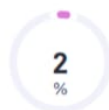


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